Does Feeling Powerful Make Us Overconfident?

A demo experiment and analysis pipeline · Nico Scholtis · October 2026

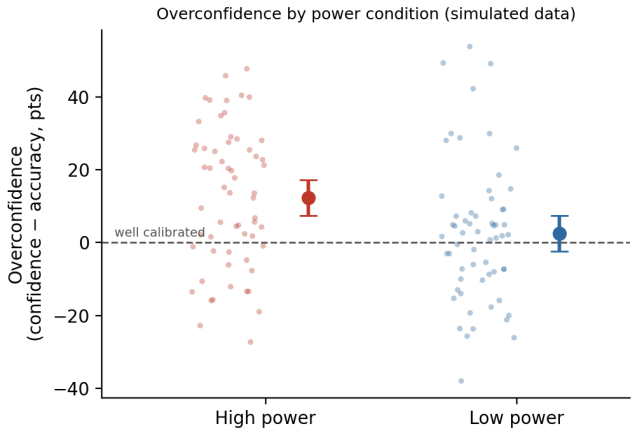
> **Note:** this is a learning exercise. The experiment is fully functional, but the results below come from **simulated data** (N = 120), generated to build and test the analysis before any real data collection. They are not empirical findings.

RESEARCH QUESTION

Research on the psychology of power suggests that holding power changes how people perceive themselves and others, and that overconfidence can spread and persist in groups. This demo asks a simple question: **does recalling an experience of power increase overconfidence in one's own judgments?**

DESIGN

- **Manipulation** (between-subjects): episodic recall of a time one had power over someone (*high power*) vs. someone had power over oneself (*low power*) (Galinsky, Gruenfeld & Magee, 2003).
- **Task:** 10 two-alternative general-knowledge questions, each followed by a confidence rating (50% = guessing, 100% = certain).
- **DV:** overconfidence = mean confidence − % correct (in points; 0 = well calibrated).
- **Checks:** felt power (1–7); accuracy, to rule out a knowledge difference.
- Built in **jsPsych** (runs in any browser); analysis in **Python** (pandas, SciPy), with a fixed random seed for reproducibility.



RESULTS (SIMULATED)

Measure	High power M (SD)	Low power M (SD)	Test
Felt power (1–7)	4.90 (1.37)	3.33 (1.13)	$t(114) = 6.82$, $p < .001$, $d = 1.25$
Accuracy (%)	66.2 (17.5)	66.8 (18.5)	$t(118) = -0.20$, $p = .839$
Overconfidence (pts)	**12.4** (19.0)	**2.6** (19.2)	$t(118) = 2.81$, $p = .006$, $d = 0.51$

The manipulation check worked, and there was no evidence of an accuracy difference between conditions. Participants were overconfident overall (M = 7.5 pts, $t(119) = 4.16$, $p < .001$), and those in the high-power condition were more overconfident (difference = 9.8 pts, 95% CI [2.9, 16.7], Welch's *t*-test). Because the data are simulated, these numbers only show that the pipeline works end to end.

LIMITATIONS & NEXT STEPS

- The simulation assumed a small true effect (5 confidence points, $d \approx 0.23$). With 60 per condition, a Monte Carlo check shows only ~24% power, so the significant result here is partly a lucky draw. A real study would need ≈ 290 per condition for 80% power, and a pre-registered sample size.
- Recall manipulations can be weak online; consider attention and writing-quality checks.
- Possible extension: a social condition to test whether overconfidence spreads from high-power partners.

HOW I USED CLAUDE CODE

- I described the study in plain language; Claude Code wrote the jsPsych experiment, the simulation and the analysis scripts.
- I ran the experiment myself as a participant, checked the data and figures, and asked Claude to justify each statistical choice (e.g., Welch vs. Student's *t*).
- Takeaway: AI agents can greatly speed up lab work, but the researcher must still check the design, the code and the interpretation.

Materials: experiment (`experiment/index.html`), simulation and analysis scripts (`analysis/`), data (`data/`). Reference: Galinsky, A. D., Gruenfeld, D. H., & Magee, J. C. (2003). From power to action. *JPSP, 85*(3), 453–466.